

NETWORKS AND COMPLEXITY

Solution 14-3

*This is an example solution from the forthcoming book *Networks and Complexity*.*

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 14.3: Stability of the SIS system [Lvl. 2]

Compute the non-trivial steady state of the SIS model

$$\dot{I} = p(N - I)I - rI$$

and determine its stability.

Solution

We already know that the non-trivial steady state is at

$$I^* = N - \frac{r}{p} \tag{1}$$

We now compute the derivative

$$\lambda = \left. \frac{\partial \dot{I}}{\partial I} \right|_* = pN - 2pI^* - r. \tag{2}$$

Substituting the steady state we find

$$\lambda = pN - 2p(N - r/p) - r \tag{3}$$

$$= -pN + r \tag{4}$$

So the non-trivial steady state is stable when $r > pN$ and unstable otherwise. So, the non-trivial state is stable when the trivial steady state is unstable and, vice versa. At $r = pN$ the two states meet and exchange their stability.